Developer laptop (Windows 11 / Linux, 4–16 GB RAM)

**Web browser**
**[Chrome / Edge]**
**Grafana :3000, BuildSim :9090,**
**occupancy UI :8081**

*127.0.0.1 port maps only*

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Docker Engine (Docker Desktop VM, 8 GB, 12 vCPU)

Compose network d7065e_default (service-name DNS)

**Shared services**
**[sim, dec, metrics-exporter]**
**d7065e-python image**

**Per-room services**
**[occ, sensor, sensor-co2, hvac, vent] × N**
**d7065e-python image**

*REST 9090 / 8081*

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*MQTT 1883*

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**Course-provided images**
**[buildsim, occupancysim]**
**Go binaries**

**Infrastructure**
**[mosquitto, prometheus, grafana]**
**upstream images**

Real deployment (not built)

**Edge gateway in the building**
**sensors, actuators, decision maker**

*off the control path*

**Cloud / campus server**
**dashboard history, tuning studies**